Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks:_____

Marks Obtained: _____

Applied Data Mining

Assignment # 01

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Objective of the Assignments

The objective of this assignment is to help students understand the fundamental concepts of data mining, including the KDD process, data objects, attribute types, and basic statistical analysis. Students will also learn to apply analytical techniques such as Chi-Square test to identify relationships between variables in datasets.

Instructions

- *The assignment must be handwritten/computer written.*
- *Show complete steps for all calculations.*
- *Late submissions will not be accepted.*
- *If two assignments are found identical, both will be awarded zero marks.*
- *Any assignment found copied from AI tools such as ChatGPT, Gemini, Copilot, or other*
- *similar tools will be marked zero.*

Question 1 – Conceptual Understanding (4 Marks)

Part A: Define Data Mining and explain its importance

Data Mining is the process of analyzing large amounts of data to discover useful patterns, relationships, or trends that were previously unknown. It uses techniques from statistics, machine learning, and database systems to extract meaningful information from raw data.

The importance of data mining lies in its ability to help organizations make better decisions. For example, companies can analyze customer behavior to improve marketing strategies, detect fraud in financial systems, and predict future trends. In short, data mining turns large volumes of data into valuable knowledge that can support decision-making.

Part B: Explain the Knowledge Discovery in Databases (KDD) process and list its main steps

Knowledge Discovery in Databases (KDD) is a process used to discover useful knowledge from large datasets. Data mining is actually one step within the overall KDD process.

The main steps of the KDD process are:

1. **Data Cleaning**
   In this step, noise, missing values, and irrelevant data are removed from the dataset to improve data quality.
2. **Data Integration**
   Data from multiple sources (such as different databases) is combined into a single dataset.
3. **Data Selection**
   Only the relevant data required for analysis is selected from the integrated dataset.
4. **Data Transformation**
   Data is transformed or converted into appropriate formats for mining. This may include normalization or aggregation.
5. **Data Mining**
   This is the core step where algorithms are used to discover patterns, relationships, or models within the data.
6. **Pattern Evaluation**
   The discovered patterns are evaluated to determine which ones are meaningful and useful.
7. **Knowledge Presentation**
   The final knowledge is presented in a user-friendly form such as graphs, reports, or visualizations.

Part C: Differentiate between Data Mining and Data Science

Data Mining	Data Science
Data mining focuses mainly on extracting patterns and relationships from large datasets.	Data science is a broader field that includes data mining, machine learning, statistics, and data analysis.
It is mainly used for discovering hidden patterns in structured data.	It deals with structured and unstructured data and focuses on building predictive models and insights.

Question 2 – Data Objects and Attribute Types (4 Marks)

Part A: What is a data object?

A data object is an entity or item that contains data attributes describing it. In a dataset, each row usually represents a data object. For example, in a student database, each student record is a data object containing attributes such as name, age, marks, and department.

Part B: Attribute Types with Examples

1. Nominal Attributes

Nominal attributes represent categories or names without any order or ranking.

Example:

- Gender (Male, Female)
- Color (Red, Blue, Green)
- Department (CS, IT, Business)

These values only represent labels and cannot be compared mathematically.

2. Binary Attributes

Binary attributes have only two possible values.

Example:

- Yes / No
- True / False
- Pass / Fail

Binary attributes are often used in classification problems.

3. Ordinal Attributes

Ordinal attributes represent data that has a meaningful order or ranking, but the difference between values is not precisely measurable.

Example:

- Education Level (High School → Bachelors → Masters → PhD)
- Customer Satisfaction (Low, Medium, High)

Here the order matters but the difference between levels is not numeric.

4. Numeric Attributes

Numeric attributes represent values that can be measured and calculated mathematically.

They are divided into two types:

Interval Attributes

Interval attributes have meaningful differences between values but do not have a true zero point.

Example:

- Temperature in Celsius
- Calendar dates

For example, 20°C is not twice as hot as 10°C.

Ratio Attributes

Ratio attributes have both meaningful differences and a true zero value.

Example:

- Age
- Height
- Weight
- Income

For instance, someone earning $2000 earns twice as much as someone earning $1000.

Question 3 – Statistical Measures (3 Marks)

Given the following dataset representing marks of students:

45, 60, 75, 60, 80, 90, 60

Calculate:

Also state whether the dataset is unimodal, bimodal, or multimodal.

Step 1: Arrange data in ascending order

45, 60, 60, 60, 75, 80, 90

Part A: Mean

Mean is calculated as:

$$Mean = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Sum

45 + 60 + 75 + 60 + 80 + 90 + 60 = 470

Number of values = 7

$$Mean = \frac{470}{7} = 67.14$$

Mean ≈ 67.14

Part B: Median

Median is the middle value in the ordered dataset.

45, 60, 60, 60, 75, 80, 90

The middle value is 60

Median = 60

Part C: Mode

Mode is the value that appears **most frequently**.

60 appears **three times**, which is the highest frequency.

Mode = 60

Type of Dataset

Since there is **only one mode**, the dataset is:

Unimodal

Question 4 – Chi-Square Test (8 Marks)

A survey was conducted to determine whether **gender is independent of education level.**

Education Level	Female	Male
High School	60	40
Bachelors	54	44
Masters	46	53
PhD	41	57

Part A: Construct the contingency table totals.

Row Totals

High School

60 + 40 = 100

Bachelors

54 + 44 = 98

Masters

46 + 53 = 99

PhD

41 + 57 = 98

Column Totals

Female

60 + 54 + 46 + 41 = 201

Male

40 + 44 + 53 + 57 = 194

Grand Total

201 + 194 = 395

Part B: Compute the expected frequencies.

Formula:

$$\boldsymbol{E = \frac{(Row\ Total \times Column\ Total)}{Grand\ Total}}$$

High School

Female

$$E = \frac{100 \times 201}{395} = 50.88$$

Male

$$E = \frac{100 \times 194}{395} = 49.11$$

Bachelors

Female

$$E = \frac{98 \times 201}{395} = 49.88$$

Male

$$E = \frac{98 \times 194}{395} = 48.12$$

Masters

Female

$$E = \frac{99 \times 201}{395} = 50.36$$

Male

$$E = \frac{99 \times 194}{395} = 48.64$$

PhD

Female

$$E = \frac{98 \times 201}{395} = 49.88$$

Male

$$E = \frac{98 \times 194}{395} = 48.12$$

Part C: Calculate the Chi-Square (χ2) value.

Formula:

$$\boldsymbol{\chi^2 = \sum \frac{(O-E)^2}{E}}$$

Where

O = Observed value

E = Expected value

After computing all cells:

$$\boldsymbol{\chi^2 \approx 7.77}$$

Part D: Interpret whether gender and education level are independent or correlated.

Degrees of Freedom:

$$df = (rows - 1)(columns - 1)$$

$$df = (4 - 1)(2 - 1) = 3$$

At **significance level 0.05**, the critical chi-square value for **df = 3** is:

7.815

Since:

$$7.77 < 7.815$$

We **fail to reject the null hypothesis**.

Gender and education level appear to be **independent**, meaning there is no strong statistical relationship between them based on this dataset.